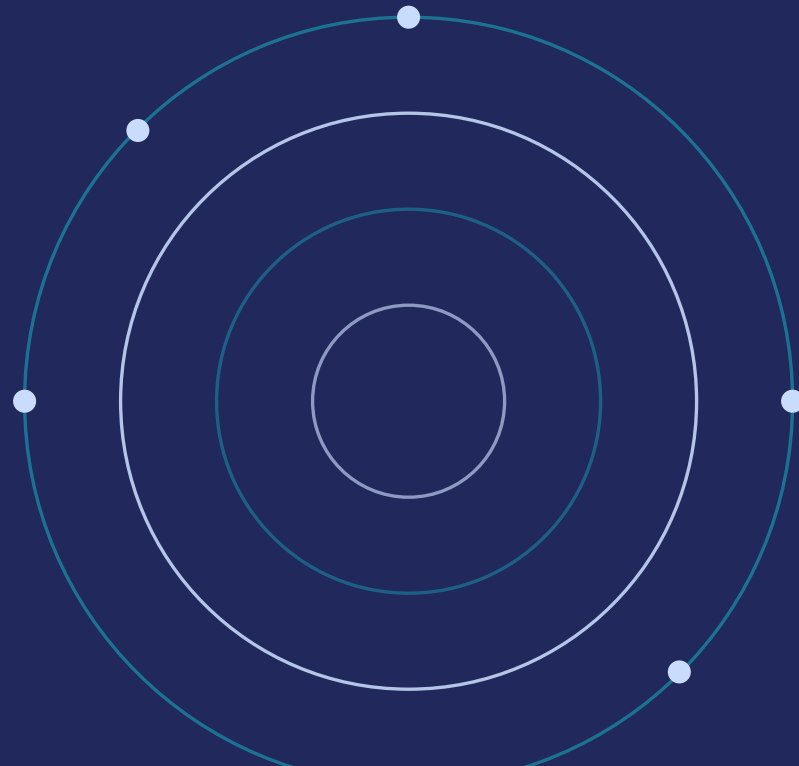


Explainability of Quantum Layers in Physics-Informed Neural Networks

A Study of Quantum-Assisted PINNs (QAPINNs) for Nonlinear and Linear PDEs

Fataneh Bakherad • Mehrdad Ghanbari Mobarakeh

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Classical PINNs Are Powerful — But Not Without Limits

CLASSICAL PINN CHALLENGES

1

Ill-conditioned optimization

The nonlinear PDE loss landscape contains poorly conditioned regions that slow or stall convergence.

2

Spectral bias

Standard networks struggle to represent high-frequency solution components, such as sharp PDE fronts.

3

Capacity vs. parameter cost

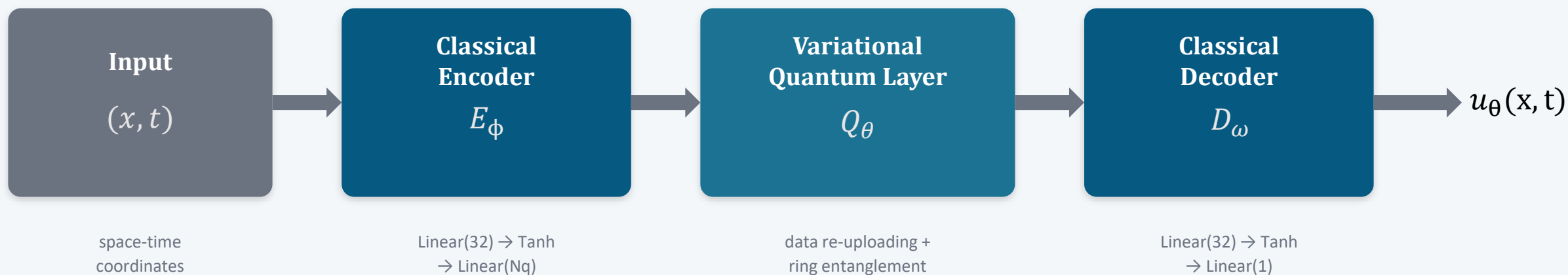
Increasing representational capacity typically means adding many more trainable parameters.

THE CENTRAL QUESTION

When, why, and how does the introduction of a quantum layer modify the learning dynamics of a physics-informed neural network?

This work does not aim to prove quantum layers are universally better — it studies the conditions under which they change optimization dynamics, parameter efficiency, and PDE approximation.

The QAPINN Pipeline



INSIDE THE QUANTUM LAYER

Data re-uploading

$RY(\pi z_q)$ re-injects the encoded coordinates at every layer, raising the effective Fourier frequency content.

Ring entanglement

CNOT(q, q+1) plus a closing CNOT(Nq-1, 0) couples all qubits with shallow circuit depth.

Z-expectation readout

$q_i = \langle \psi | Z_i | \psi \rangle \in [-1, 1]$ gives a bounded, differentiable feature vector.

Four Principles Behind the Quantum Layer

1

Continuous encoding for continuous variables

Space-time coordinates are injected through rotation angles rather than discrete basis encoding.

2

Repeated encoding for nonlinear PDEs

Data re-uploading increases the effective Fourier representation, capturing sharper solution structures.

3

Entanglement should match physical correlation

Ring entanglement exchanges information across all variables while keeping circuit depth shallow.

4

Depth should be optimized, not maximized

More layers raise expressivity but can also destabilize gradients and hurt trainability.

Two Benchmark PDEs, One Shared Methodology

Allen-Cahn Equation

$$u_t - \varepsilon^2 u_{xx} + u^3 - u = 0$$

Nonlinear · sharp interfaces

Heat Equation

$$u_t = \alpha u_{xx}$$

Linear · smooth diffusion

TRAINING CONFIGURATION

Optimizer	Adam
Initial learning rate	1×10^{-3}
Activation function	tanh
Training epochs	3,000 – 8,000
Quantum simulator	PennyLane default.qubit
Sampling method	Latin Hypercube Sampling

PHYSICS-INFORMED LOSS

L_PDE	Governing-equation residual at collocation points inside the domain
L_IC	Initial-condition points enforcing the known initial solution
L_BC	Boundary-condition points enforcing spatial constraints

Fewer Qubits Add Depth; More Qubits Add Room

Model	Qubits	Layers	Params	Rel. L2 Error	Time (s)
PINN (classical)	—	—	1185	0.0489	57.8
QAPINN	2	1	295	0.1093	276.9
QAPINN	2	2	299	0.0450	460.8
QAPINN	2	3	303	0.0475	644.5
QAPINN	6	1	563	0.00031	5925.3
QAPINN	6	2	575	0.00150	11014.6

3.1×10^{-4}

best relative L2 error — 6 qubits, 1 layer, ~160×
lower than the classical baseline

52.5%

fewer trainable parameters (563 vs 1185) while
beating the classical baseline's accuracy

Depth helped the 2-qubit circuit (1→2 layers) but then plateaued; the 6-qubit, 1-layer circuit was best overall — adding a second layer there made things worse. Expressivity gained from more qubits outperformed expressivity gained from more depth.

Here, Depth Is What Pays Off

Model	Qubits	Layers	Params	Rel. L2 Error	Time (s)
PINN (classical)	—	—	3297	0.00214	97.2
QAPINN	2	1	10	0.0911	624.8
QAPINN	4	1	18	0.1623	2045.6
QAPINN	2	3	26	0.0100	1621.9
QAPINN	4	3	50	0.00451	5349.0

18×

fewer parameters at 4 qubits, 3 layers (50) than
the classical PINN (3297)

0.133

extrapolation L2 error for the 2-qubit, 1-layer
QAPINN — vs. 0.0212 for the classical PINN

Unlike Allen–Cahn, accuracy here scaled with circuit depth (L2: 0.091→0.010 for 2 qubits; 0.162→0.0045 for 4 qubits). But better interpolation did not carry over to extrapolation — all QAPINN variants generalized worse in time than the classical baseline.

The Optimal Circuit Is Problem-Dependent

Characteristic	Allen–Cahn	Heat Equation
PDE type	Nonlinear reaction-diffusion	Linear diffusion
Solution structure	Sharp interfaces	Smooth evolution
Most effective lever	Increase qubit count	Increase circuit depth
Main challenge	Representing nonlinear features	Approximation accuracy

TAKEAWAY

Sharp, high-frequency solutions reward a larger quantum feature space (more qubits); smooth, low-frequency solutions reward more repeated nonlinear transformations (more depth). Circuit design should follow the PDE's spectral character — not a one-size-fits-all template.

What the Quantum Layer Gains — and Costs

Gained

- + Fewer trainable parameters for comparable or better accuracy
- + A genuinely nonlinear feature transform via quantum evolution
- + Extra design flexibility through encoding and measurement choices
- + A compact representation for problems needing complex feature spaces

Lost / At Risk

- Substantially longer training time from quantum simulation (up to ~100× the classical run)
- Strong sensitivity to circuit architecture and depth choices
- Deeper circuits can reduce trainability rather than improve it
- No guaranteed improvement in extrapolation / generalization

A Recommended Workflow for Problem-Specific QAPINNs

1

Analyze PDE properties: linear vs. nonlinear, smoothness, sharp gradients

2

Size the circuit: 2–4 qubits for smooth PDEs, 4–8 for strongly nonlinear ones

3

Start shallow: begin with a single circuit layer ($L = 1$)

4

Increase qubits before increasing depth

5

Evaluate PDE residual, relative L2 error, gradient stability, and cost

6

Add depth only if extra expressivity is still required

CONCLUSION

The quantum layer is best understood as an alternative, problem-dependent feature map — not a drop-in replacement for classical layers. Its value shows up as parameter efficiency and representational change, matched by real costs in simulation time and trainability.